

# Species Distribution Modeling: Crash Course Part 1

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# Overview

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What drives a species distribution?

The Niche Concept

Broad SDM workflow

SDM Assumptions & Violations

What questions can we ask?

# What are Species Distribution Models?

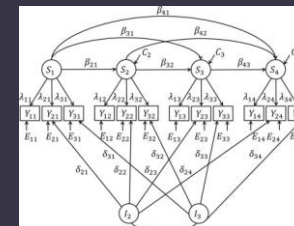
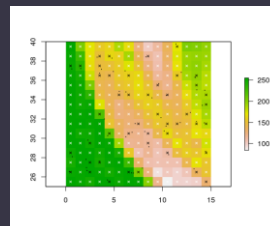
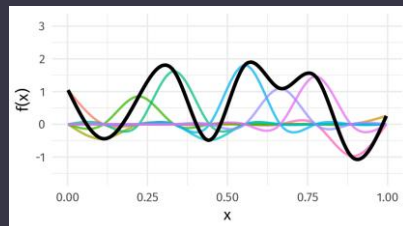
Species Distribution Models (SDMs): “Numerical tools that combine observations of species occurrence or abundance with environmental estimates,” in order to project what areas in space are suitable for species persistence.

- Also referred to as ecological niche models (ENMs) or habitat suitability models

Correlative SDMs: Produce function that encapsulates relationship between environmental variables and species presence

Pro: So many models, minimal knowledge of physiology req.

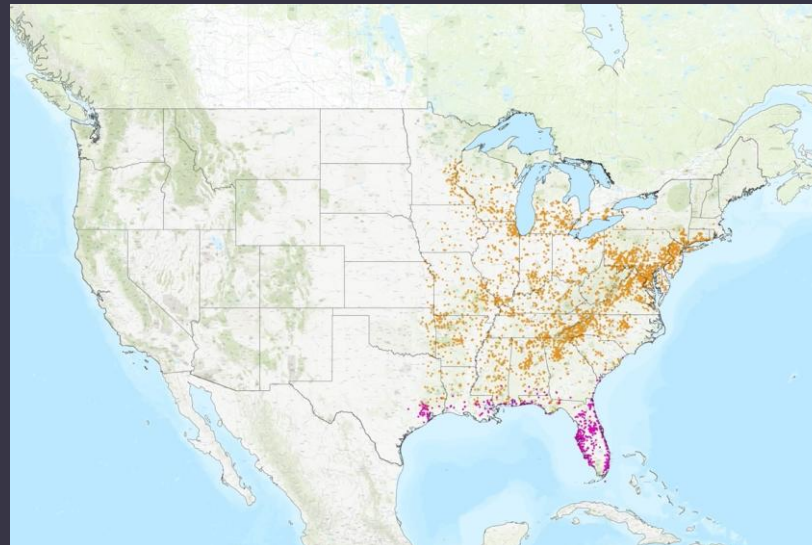
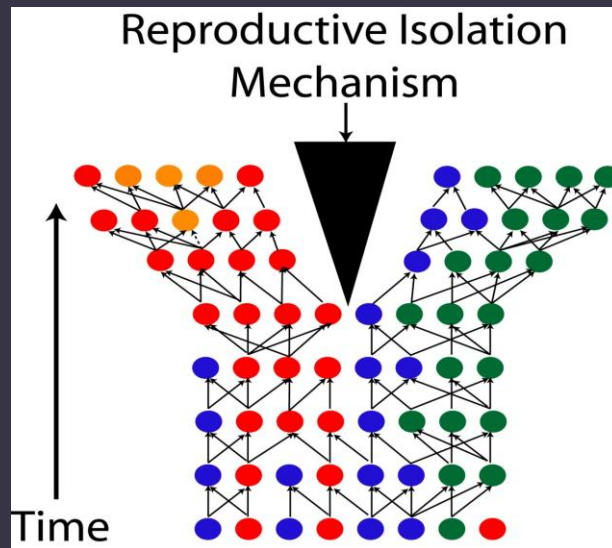
Con: So many models, use of physiological data minimized



# What Drives a Species Distribution?

To persist in a particular place a species must

1. Have *evolved* to survive in that place (or reach a suitable place through dispersal)
2. Tolerate the *biotic* interactions in that place
3. Be able to maintain population growth in the *abiotic* conditions of that place



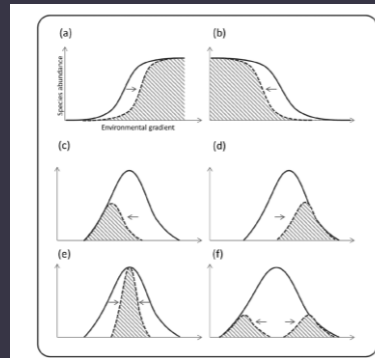
# Conditions for living in a given place

## Evolved (or dispersed) there

- Dispersal mechanisms available
- Via **allopatry** (due to geographic barrier)
  - River, mountain range, ocean, desert, drainage basin
- Via **sympatry** (other than geographic barrier)
  - Niche partitioning, chromosomal incompatibility, temporal reproductive isolation

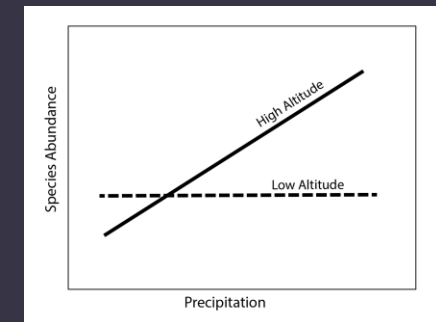
## Biotic tolerance

- Trophic interactions
  - Predator / Prey
- Symbiotic interactions
  - Pollinator / Pollinatee
- Competitive interactions
  - Competitive exclusion

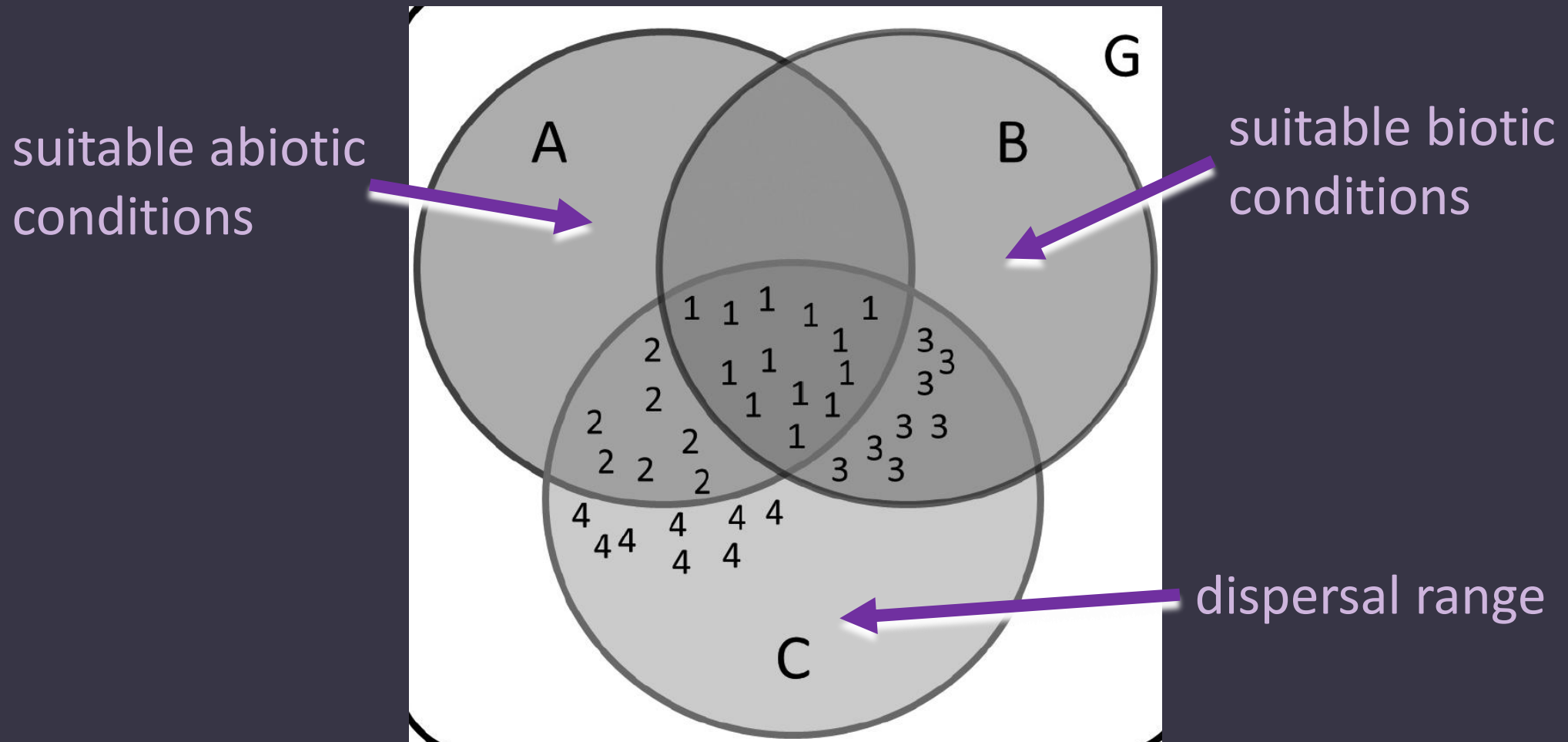


## Abiotic tolerance

- Temperature
- Precipitation
- Radiation
- Seasonality
- Soil Minerals
- Etc

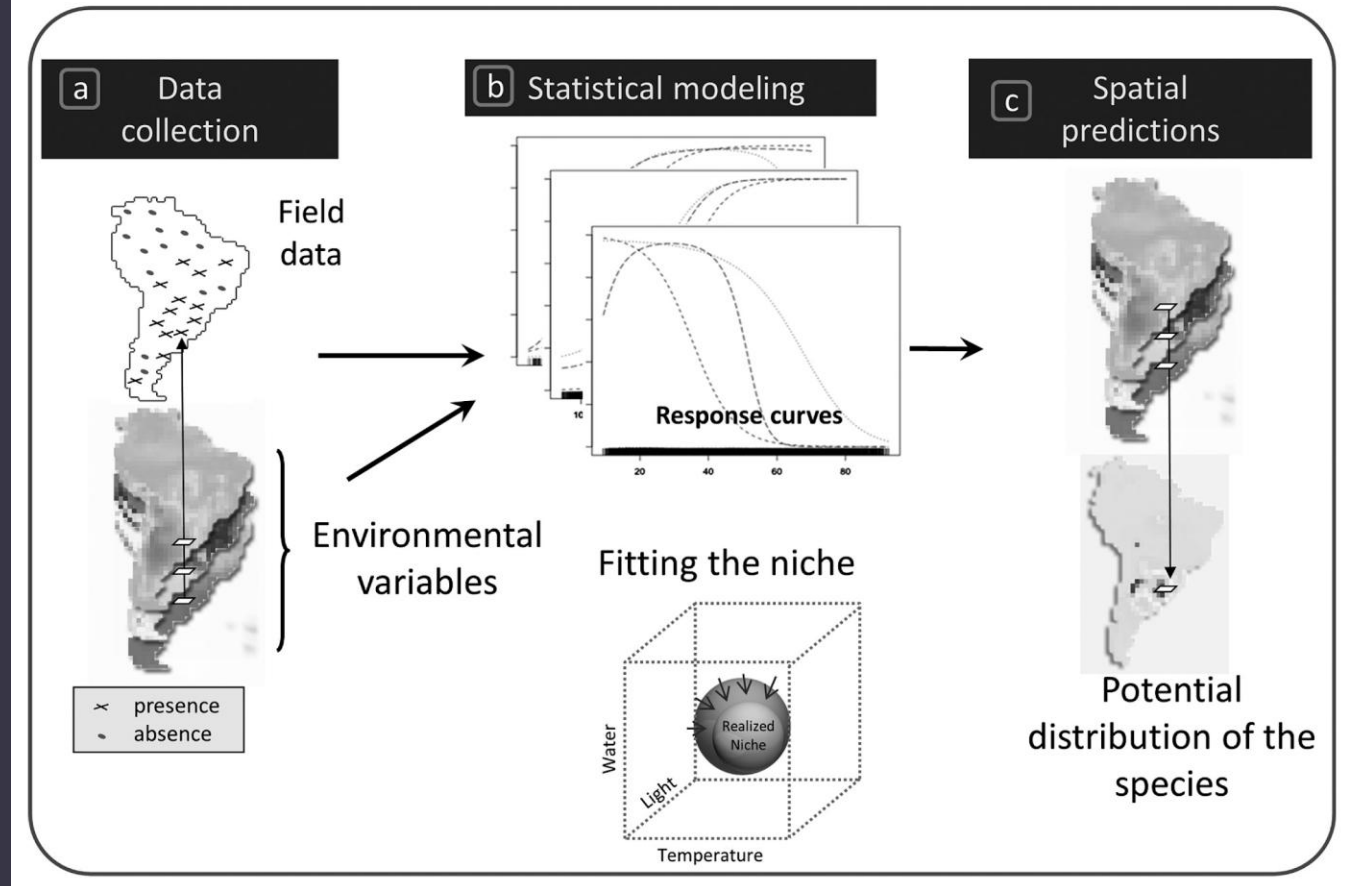


**Realized Niche:** The fundamental niche once it has been constrained by dispersal capability, biotic filtering, and the range of environmental factors that are **ACTUALLY** available to species.

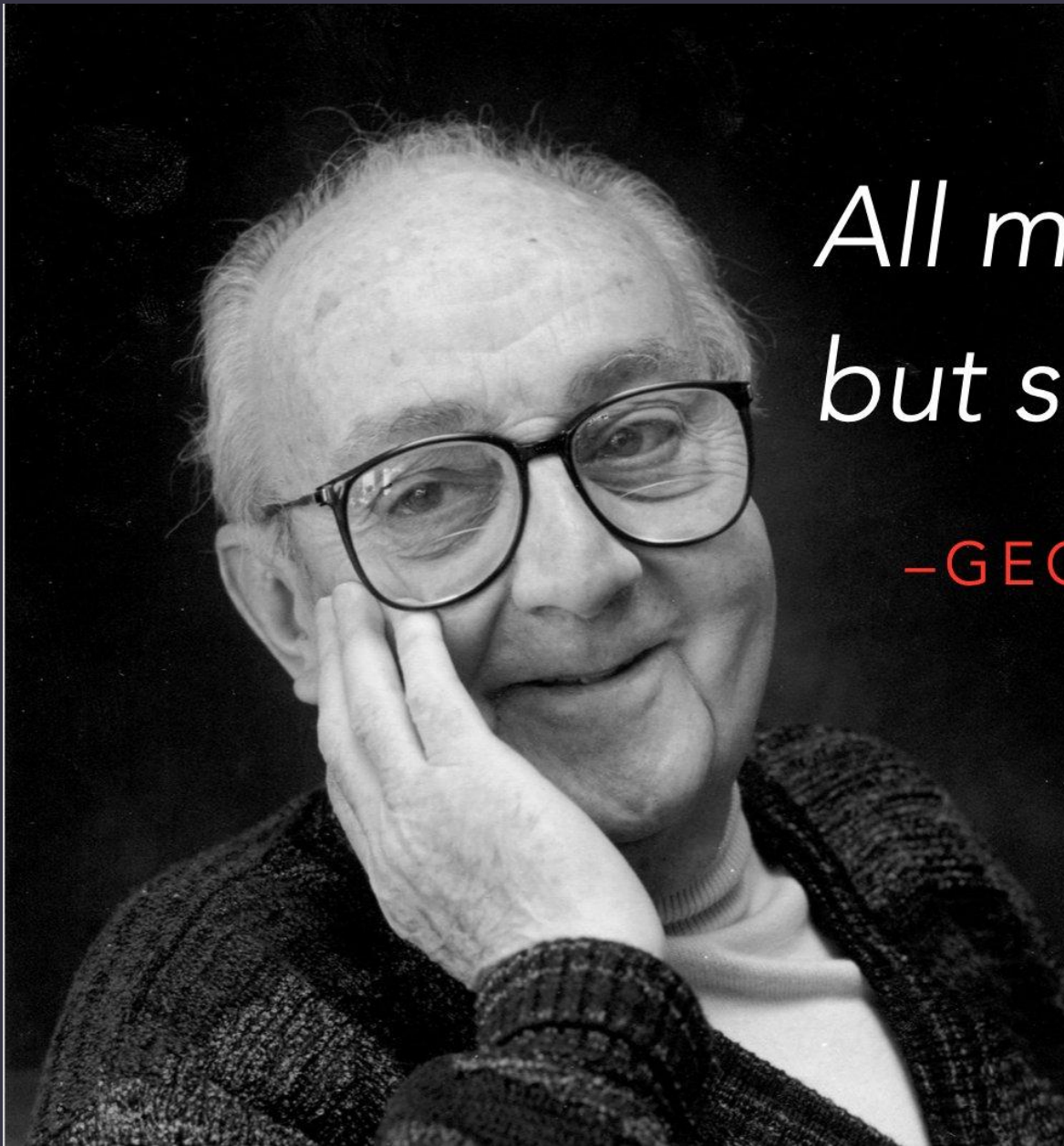


# SDM: A Coarse Overview

1. Collect occurrence and predictor data
2. Fit/tune models
3. Evaluate model fit
4. Project model fit into space







*All models are wrong,  
but some are useful.*

—GEORGE BOX, UW-MADISON



# SDM: A Finer Overview

**Data:** Occurrence/Abundance,  
Background Points, Predictor Variables

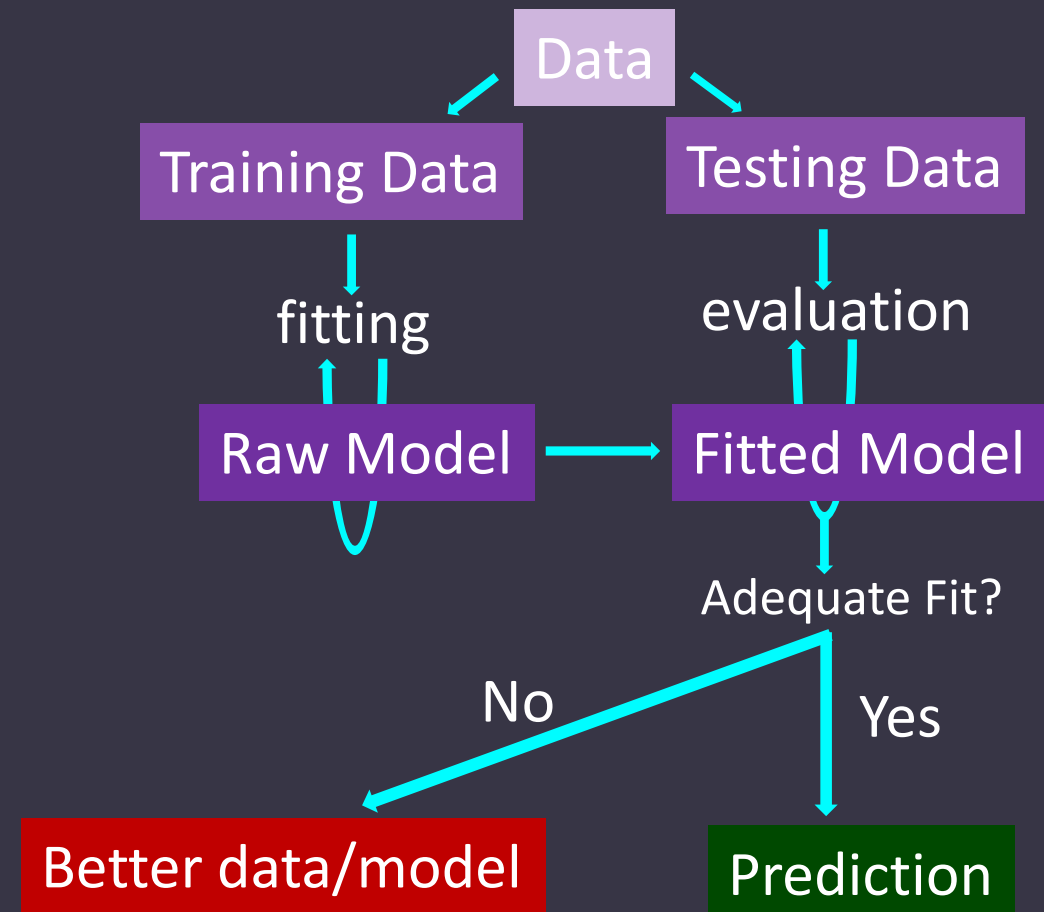
**Train/Test Set:** Normally 80/20 or 75/25

**Fitting:** Model tuning (RM, Feature  
Class, LR, TC)

**Evaluation:** AUC-ROC, TSS, Max Kappa,  
Boyce Index

**Good Fit?:** Proceed to spatial projection

**Bad Fit?:** Go back to step 1 and re-  
evaluate



# Data: Occurrences

Where is a species known to occur?

Surveys, collections, databases, literature

Different than abundance data

- # individuals at a site, not commonly available

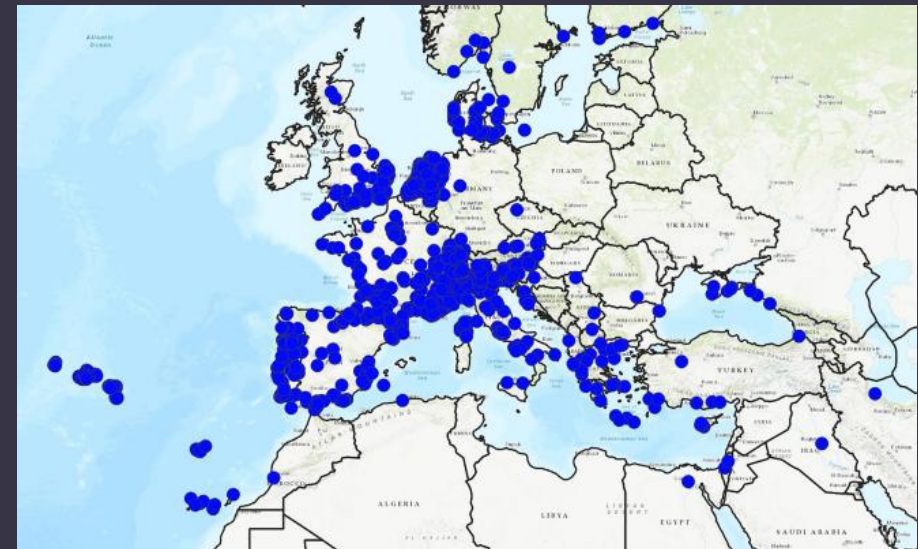
Records need to be cleaned

- Flip-flop Lat/Lon, terrestrial species in ocean, lat/lon corresponds to museum across the plant from known range, etc.

Some methods require absence data

- More appropriately: pseudo-absence data or background data

Longitude/Latitude coordinates of individuals (*Chrysodeixis chalcites* (Esper))



# Data: Background/Pseudoabsence Points

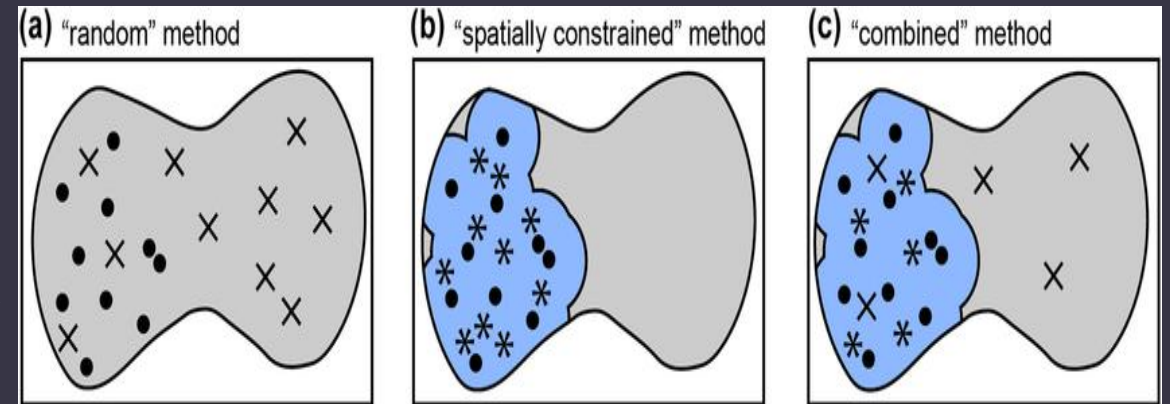
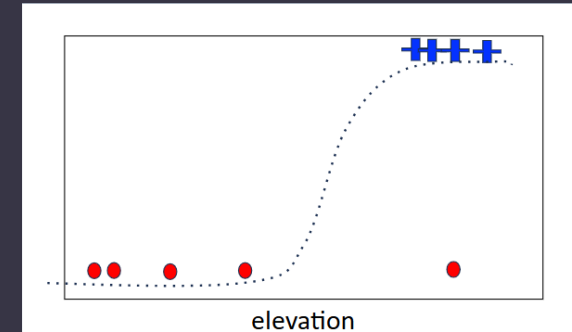
Where is a species not known to occur?

Absence/Pseudoabsence/Background: where a species is not known to occur

- Most repositories do not include information about where species are not found
- Many researchers don't report it anyways

Why do we need this?

- 1) Provides contrast to presence/env. correlation (samples the variation of the landscape)
- 2) Allows for model evaluation: how well does the model predict presence data compared to random data?



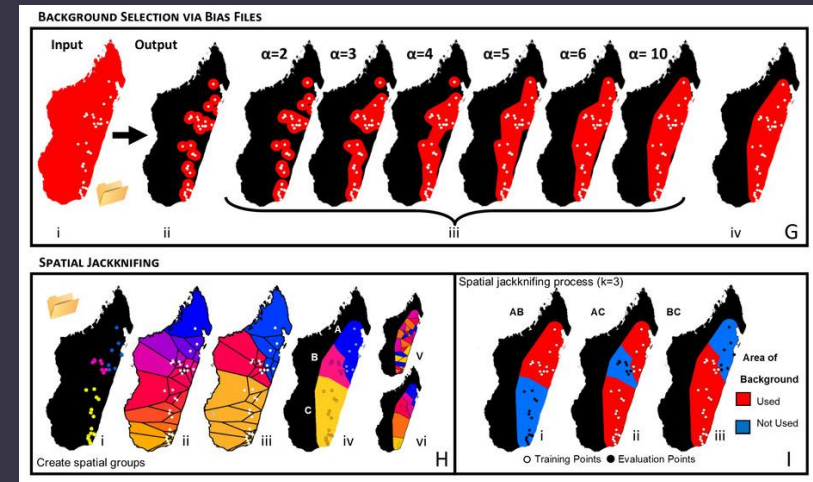
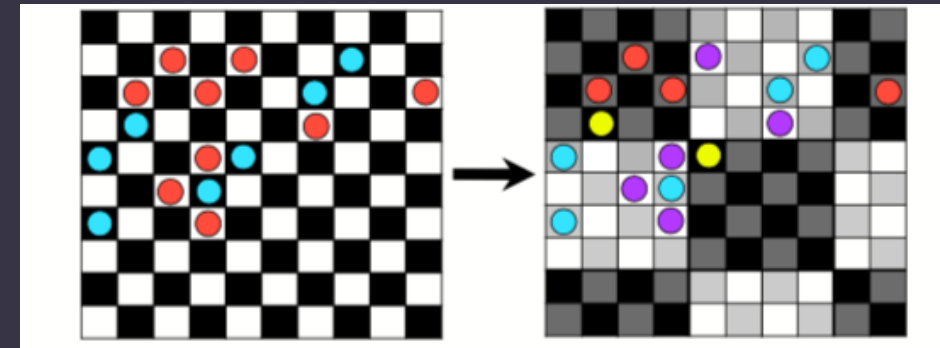
\* This topic could use its own presentation

# Data: Train / Test Partition

## Subset into training and testing

- Random subset (not based on space)
- Checkerboard 1/2 sampling (ensures a more even spatial distribution between groups)
- Cluster-based

More in-depth as part of another talk



# Data: Environmental

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What do we know about where species occur?

Weather and Climate

Solar Radiation

Soil Type

Vegetative Cover

Human Disturbance

Elevation/Slope/Aspect



# Data: Environmental

Generally stored as raster maps

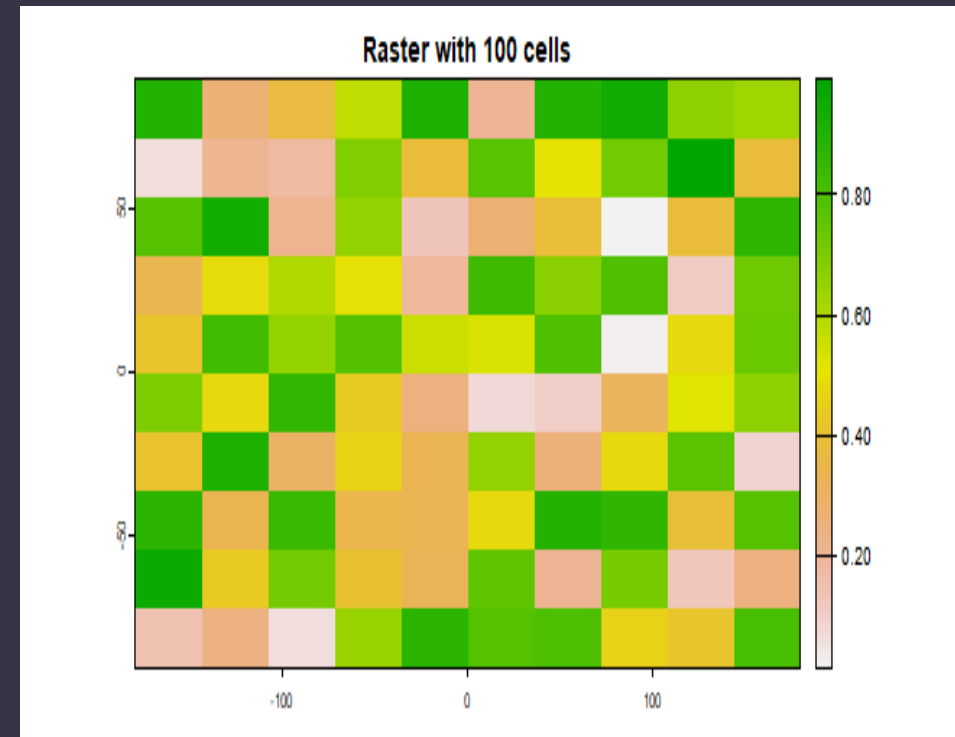
- Raster = grid of cells/pixels, each representing a real space in the world
- Pixels encode value of an environmental condition

Some predictors play a larger role than others for a given species, how can we determine which is which?

- Many approaches out there, many just try a lot of combinations

Consideration of temporal scale important

Consideration of spatial resolution important





# Data: Physiological

## What do we know about the species tolerances?

Knowing physiological limits can allow us to:

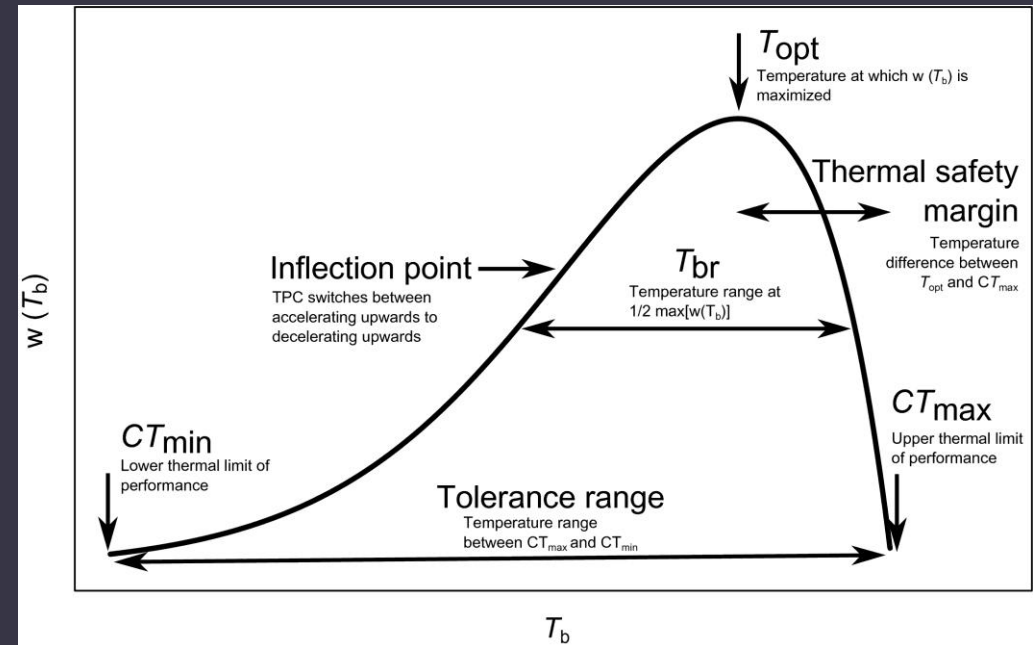
- 1) Clamp the extremes
- 2) Incorporate physiology into correlative frameworks

### Temperature-Dependent Performance

- Probably the most abundant, characterized through TPCs

### Xyz-Dependent Performance

- Salinity, humidity, etc



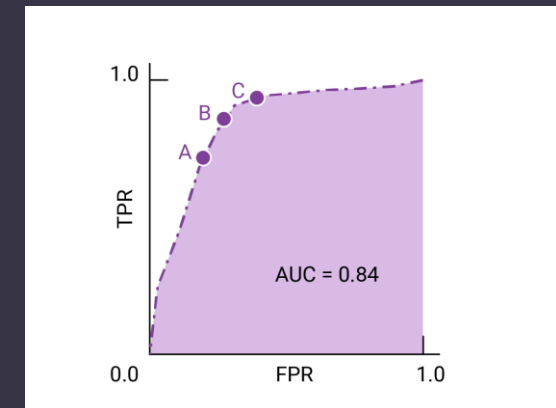
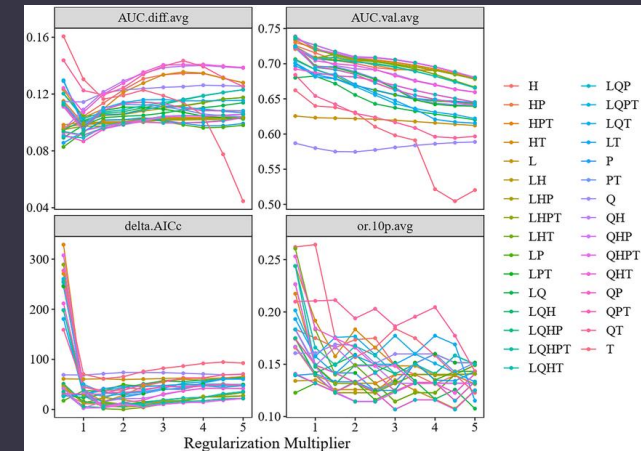
# SDM: Fit & Evaluate

**Model Selection:** Dependent on modeling type (feature class, regularization multiplier, tree complexity, learning rate, etc.). Usually based on lowest  $\Delta AICc$

**Fit evaluation:** Test the fit against the withheld testing set. Often used include AUC-ROC, TSS, Boyce Index

**Spatial Projection:** Spatial projections of SDMs are just projections of the residual fit

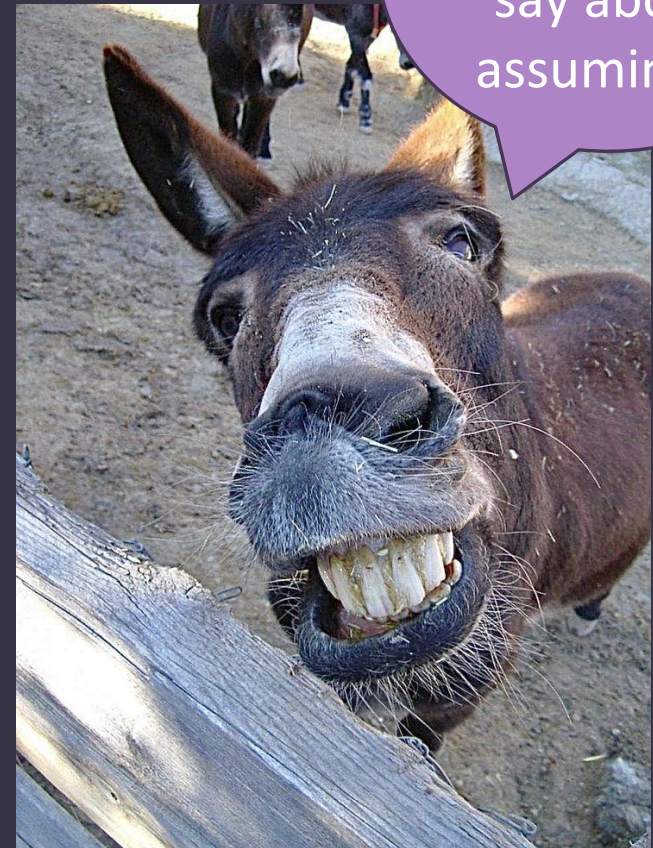
**Ensemble of Models (Optional):** Ensemble predictions of multiple models (often performance-weighted averaging)



# SDM: Assumptions

Species distribution modeling follows some general assumptions

1. Species-Environment Equilibrium
2. Availability of Important Predictor Variables
3. Appropriateness of Species Observations
4. Appropriateness of Statistical Methods
5. Occurrence Data is Unbiased
6. Occurrence Data is *not* Autocorrelated

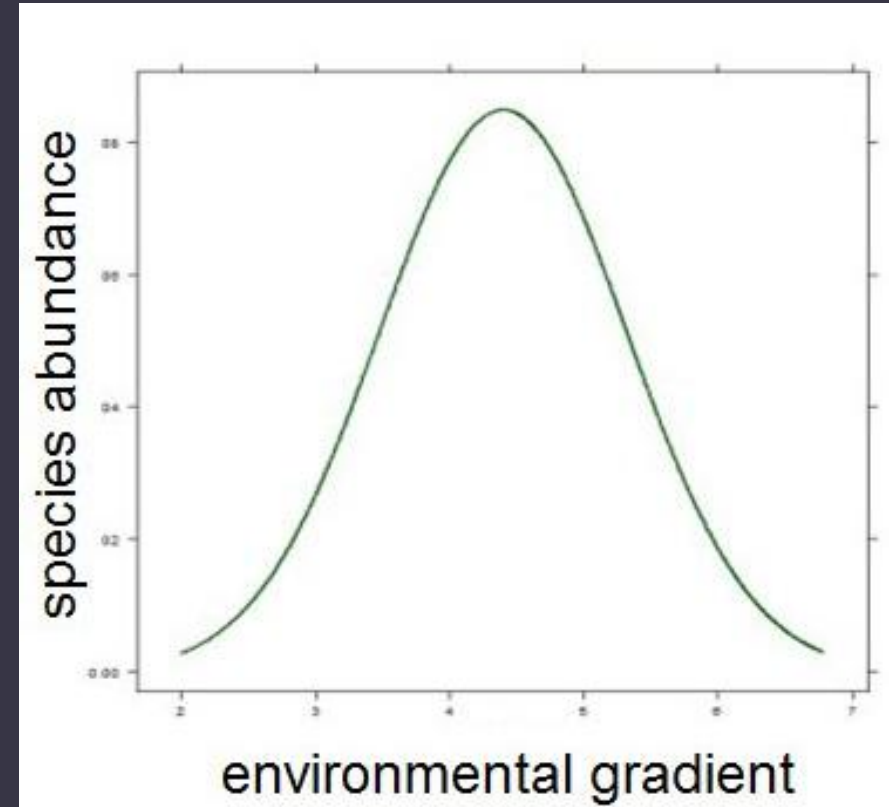


You know  
what they  
say about  
assuming...

# Assumption 1: Species-Environment Equilibrium

Sampled occurrence data captures a sample of a species realized niche

- There is a relationship between the environment & probability of occurrence
- A species fills all of the niche space available to it via dispersal
- Selection *maintains a stable niche*

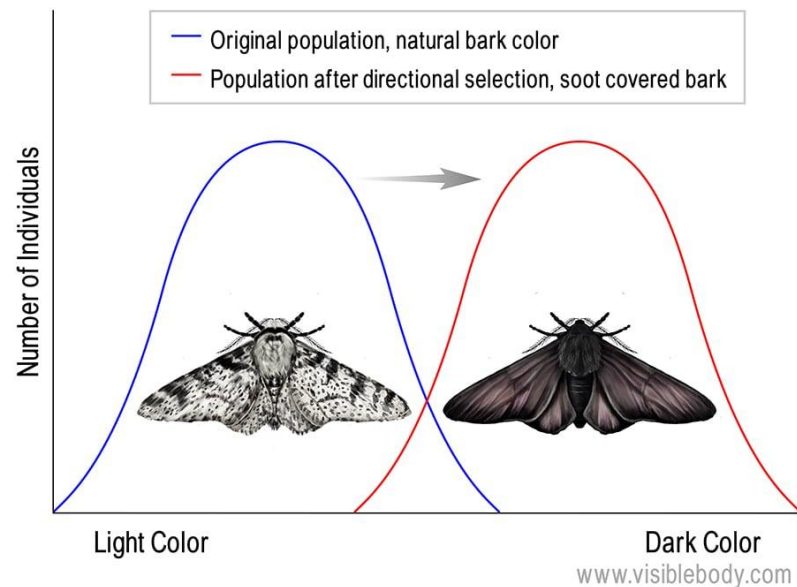


This relationship remains static

# Assumption 1: Species-Environment Equilibrium

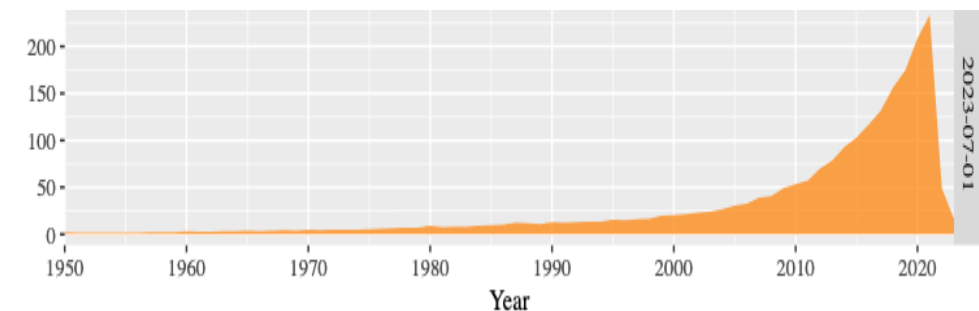
Violation: Species realized niche is changing due to selection pressures

## DIRECTIONAL SELECTION



Caveat: This change may be negligible within the time period that occurrence data has been collected

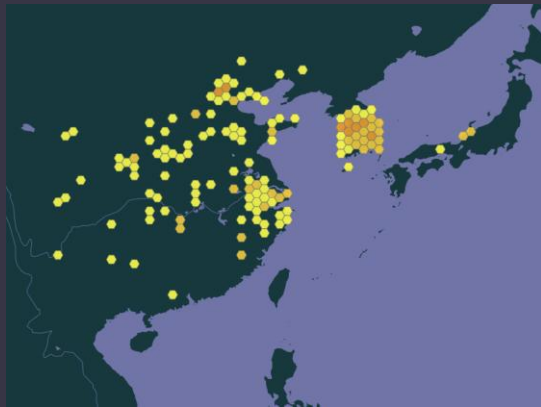
## Number of occurrences (millions) by collection date in GBIF



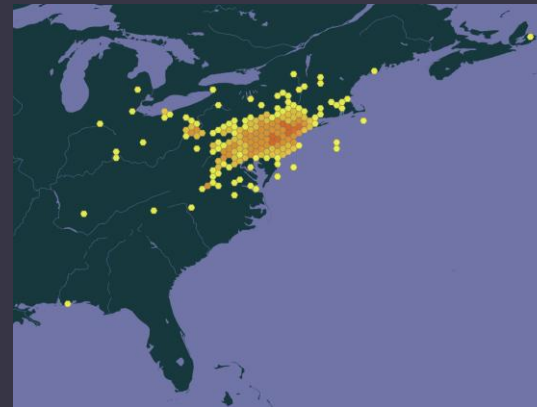
# Assumption 1: Species-Environment Equilibrium

Violation: Species realized niche is changing due to selection pressures

Invasive species: once dispersed to a new environment biotic release often allows expansion beyond native realized niche. This means they often do better in their new range due to lack of pathogens/parasites/competitors/predators



*Lycorma delicatula*: Native range



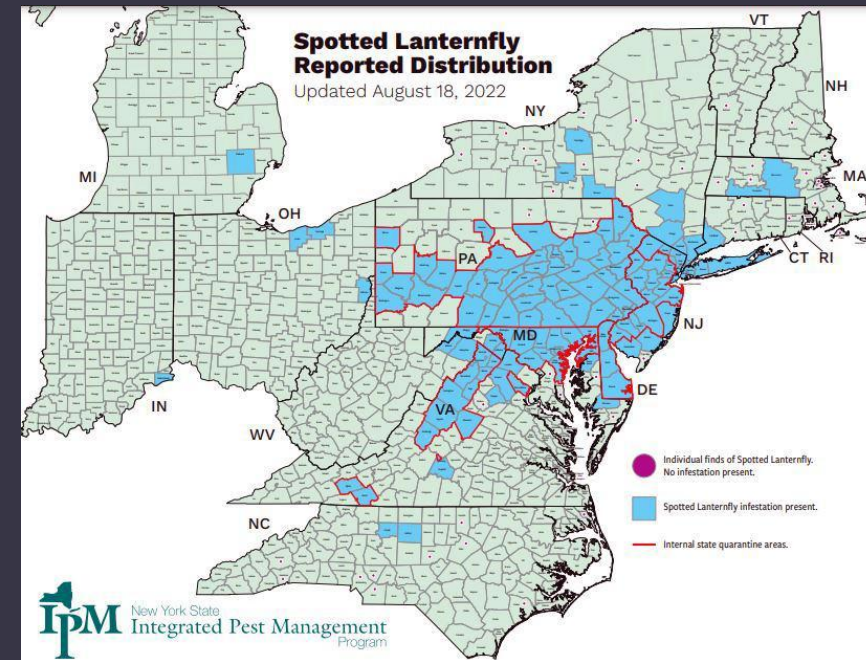
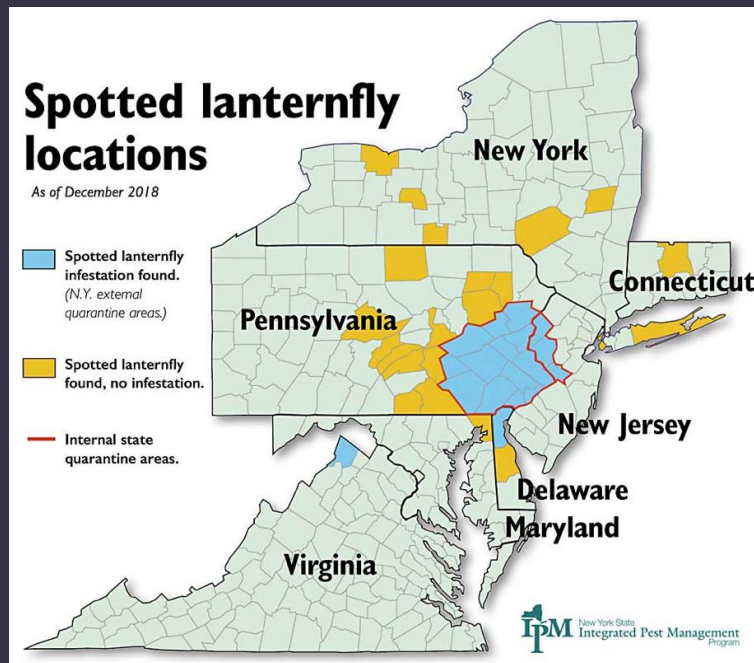
*Lycorma delicatula*: Invaded range





# Assumption 1: Species-Environment Equilibrium

Violation: Species realized niche is changing due to selection pressures



Some invasive species may not yet be in equilibrium with their new habitat.  
If that is the case, train your models using the home range only (or both)

# Assumption 2: Important Environmental Predictors Are Available

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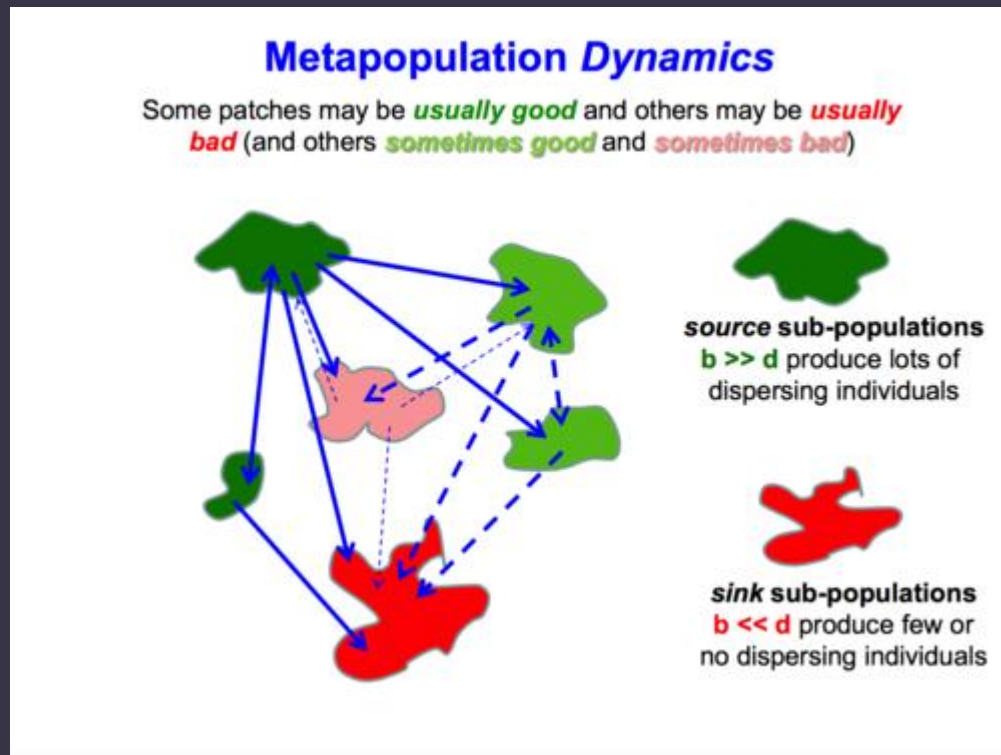
We do not always data for environmental variables that are important for a given species survival

Ex: very few relevant predictor variables (and their associated data) are available for deep water species



# Assumption 3: Appropriateness of Species Observations

Is the species able to experience positive population growth where it was found?



# Assumption 4: Appropriateness of Statistical Methods

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Is a predictor amenable to being modeled?

## Numerical Data

- % Clay
- mm precipitation in March
- Diurnal temperature range
- Units of photosynthetically active radiation per week

## Categorical Data

1. Clay
2. Sand
3. Silt
4. Loam

You can code soil categories as integers, but these are symbolic and do not have any mathematical relationship



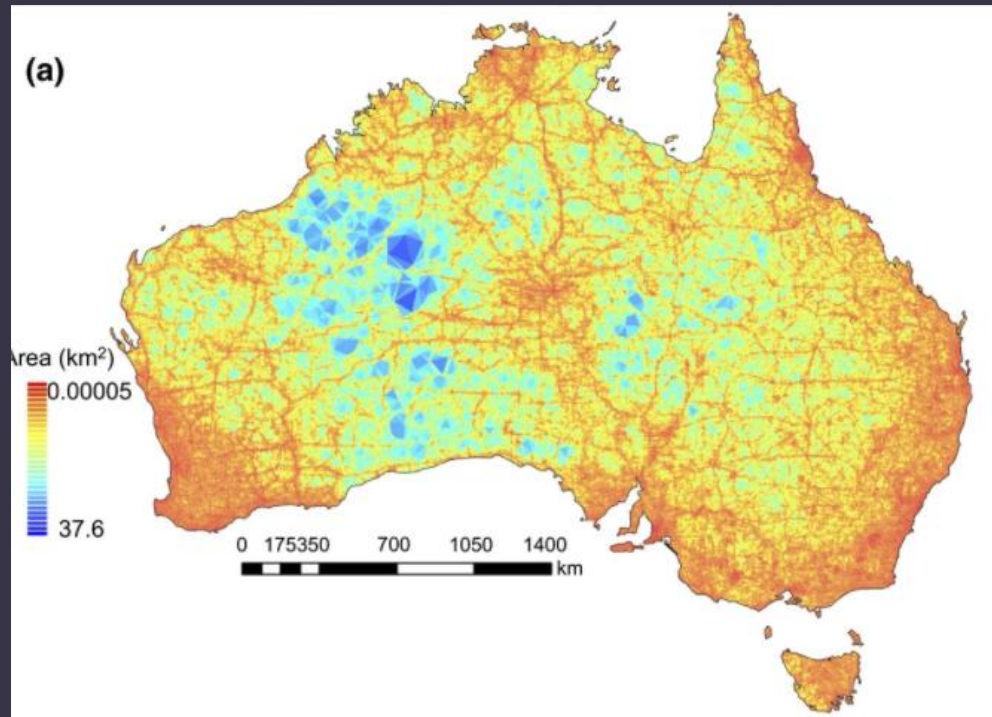
# Assumption 5: Occurrence Data is Unbiased

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People don't go out and randomly collect occurrence records

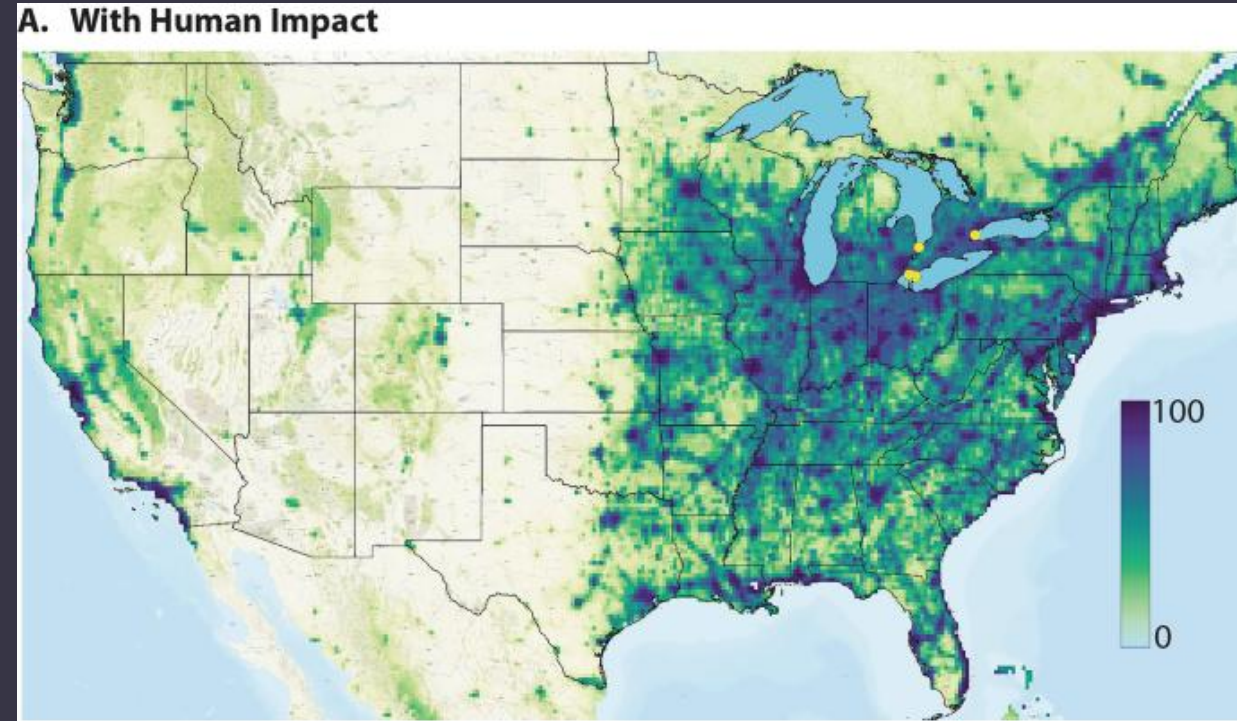
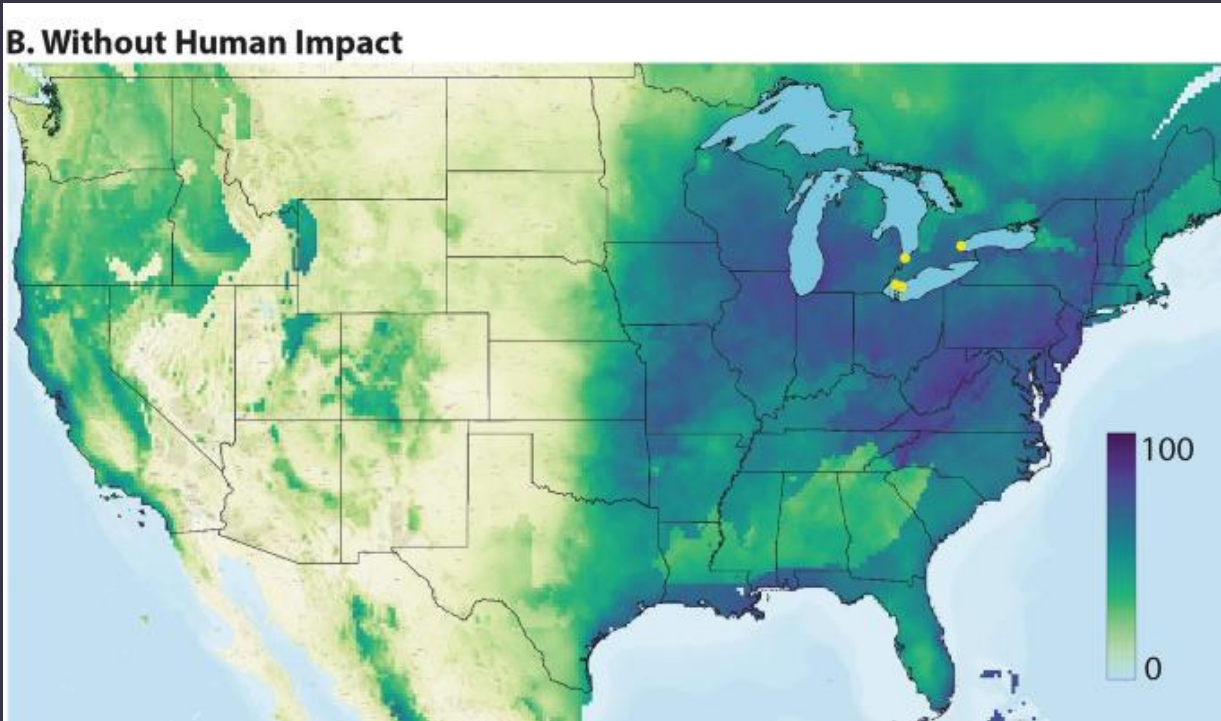
- Near roads, habitation/collection facilities
- Megafauna and flora favored in sampling record
- Fungi, micro-organisms under-represented

There are ways to combat this bias



Avg. area / Collection

# Example of Bias in Spatial Projections





# Assumption 6: Occurrence Data is not Autocorrelated

Spatial autocorrelation: the probability of a species occupying area XYZ is dependent on the probability of a species occupying a space adjacent to XYZ

Organisms are likely to live close to where they were born/hatched/budded

Species are also likely to be found where they have been found before, even under changes in condition



Species distributions are driven by previous distributions as well as niche

# SDM: Application-Based Uses

## Application-based:

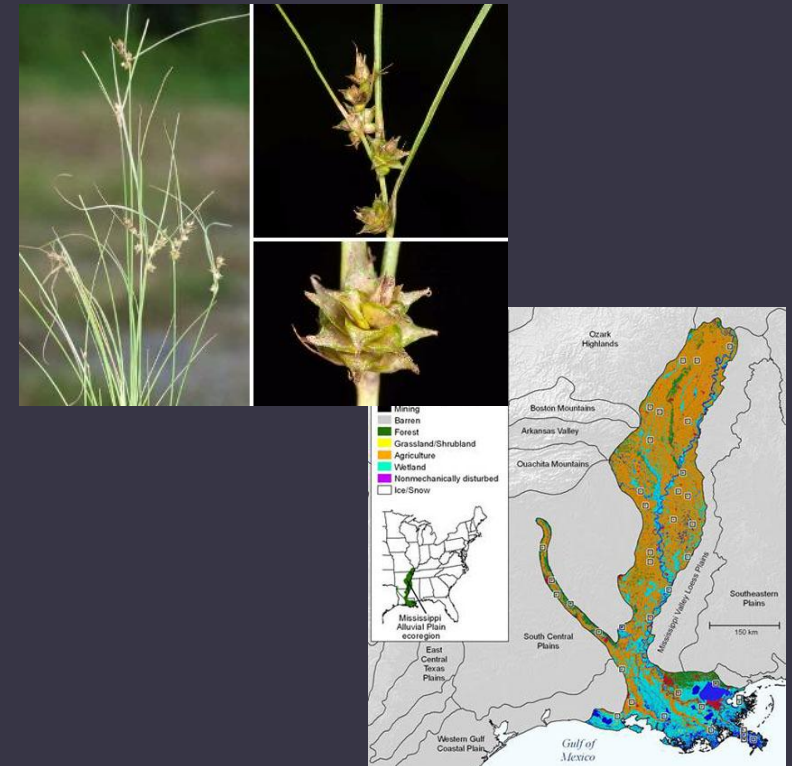
Predict changes in habitat suitability under change scenario/time

Map invasion potential

Help inform what species need more (or less) focus

Predict past distributions and shifts

Identify areas that *may* contain new populations of rare species



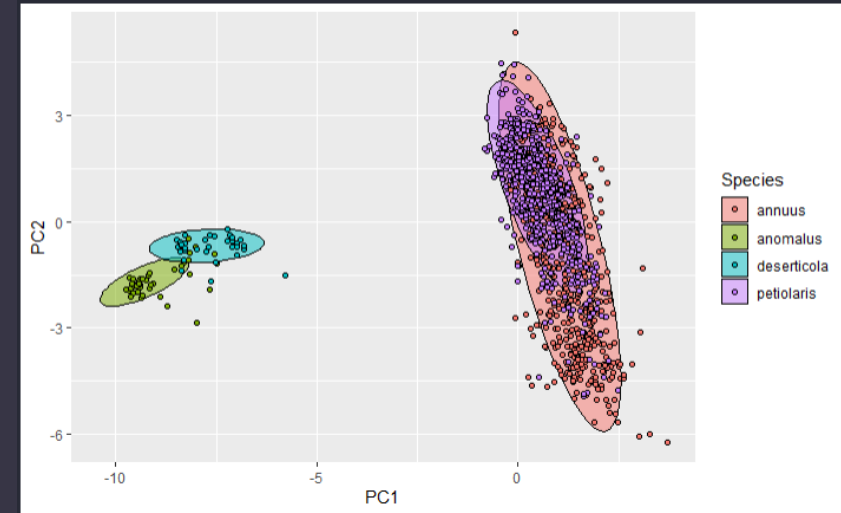
# SDM: Theory-Based Uses

## Theory-based:

Quantify the niche space of an organism/population

Identify factors that are highly associated with a given distribution

Add support for hypotheses regarding speciation



# Future Meeting Ideas

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1. Occurrence points, background points, spatial partitioning
2. Predictor data
3. Model selection
4. Evaluating SDMs
5. General workshops: I have code and exercises already written up